

DRY-RUN GUIDE | 02_maximum_sum_subarray_size_k.py

Source: Python\dsa_solutions\02_maximum_sum_subarray_size_k.py

CORE BEHAVIOR

Maintains a contiguous fixed-size window and updates its aggregate as the window moves.

REPRESENTATIVE INPUT

Array '[2,1,5,1,3,2]', k=3.

STEP TRACE

Window sums are 8, 7, 9, 6; best sum=9.

EXPECTED OUTPUT / RETURN

Maximum window sum '9'.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\dsa_solutions\02_maximum_sum_subarray_size_k.py; line numbers refer to the original file. No source was modified.

```
001 | import sys
002 |
003 |
004 | def main():
005 |     data = list(map(int, sys.stdin.read().split()))
006 |     if not data:
007 |         return
008 |     count = data[0]
009 |     numbers = data[1:1 + count]
010 |     window_size = data[1 + count]
011 |     window_sum = sum(numbers[:window_size])
012 |     best_sum = window_sum
013 |     for index in range(window_size, count):
014 |         window_sum += numbers[index] - numbers[index - window_size]
015 |         best_sum = max(best_sum, window_sum)
016 |     print(best_sum)
017 |
018 |
019 | if __name__ == "__main__":
020 |     main()
021 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.